

Set Membership Localization of Mobile Robots via Angle Measurements

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Abstract—This paper addresses the localization problem for a mobile robot navigating in an unstructured outdoor environment. A new technique is introduced, for computing an estimate of the position of the robot and the related uncertainty region, in the presence of visual angle measurements affected by bounded errors. The proposed set membership estimation procedure exploits the structure of the static set estimator, to solve recursively the dynamic localization problem.

Index Terms—Angle measurements, landmarks, localization, mobile robotics, set membership.

I. INTRODUCTION

ONE OF THE important issues for autonomous navigation in an unstructured outdoor environment is the localization problem (see [1]–[3] and references therein). Environmental conditions, sensor limitations and map inaccuracies require that the navigating robot periodically computes its coordinates in the reference system by exploiting geometric properties and measurements based on landmarks distributed in the environment. In the present paper, the relative bearing approach to the map localization problem for a mobile robot is considered. In this setting, it is assumed that the robot moves on a terrain map and is able to perform angular measurements with respect to known landmarks (see [4]–[8]). Three main sources of uncertainty characterize the localization problem in this context: a) measurement errors of the visual angles from the robot standpoint; b) landmark misidentification; c) environment map errors. The fact that the dominant component of uncertainty is of the first type, motivated the introduction of estimation approaches explicitly accounting for angle measurement errors ([5], [7], [9]). Batch or recursive techniques have been used, according to the static or dynamic setting of the position estimation problem. However, most of the results available in the literature refer to the static problem, while relatively little attention has been devoted to the dynamic problem [10], [11].

Different approaches have been adopted in the literature to tackle the localization problem. Most popular techniques rely on statistical assumptions about the error and adopt standard estimation tools to compute the robot position. For example, the triangulation problem is solved via linear least-squares algorithms in [7] and [12], in two different contexts. Extended Kalman filtering is generally adopted when facing the dynamic

localization problem (see, e.g., [10], [11], [13]). Other localization techniques based on different probabilistic paradigms, like Bayesian filtering [14] and maximum-likelihood estimation [15], have been recently introduced.

A quite different way to tackle the problem is to assume that all the uncertainty sources generate errors that are unknown but bounded. This has given rise to localization methods based on geometric properties of the measurement model [5], or on the computation of feasibility regions where the robot is known to lie, the so-called *set membership* approach [6], [16]. Properties of statistical and set-theoretic techniques for sensor data fusion have been analyzed in [17].

The contribution of the present paper is twofold. First, it introduces a new estimation technique which allows to derive an estimate of the position of the robot and the related uncertainty region, in the presence of angle measurements affected by bounded errors. No statistical assumption is made on the errors; the only assumption is that they are bounded in norm by some known quantity. The second objective is to provide a procedure which exploits the structure of the static set estimator to solve recursively the dynamic localization problem. The basic reference theory for the technical development of the paper relies on set theoretic concepts which have been recently investigated in the set membership estimation area of research (see, e.g., [18]–[20]). Although most of the general theory developed in these references refers to linear estimation problems, the specific nonlinear structure of the admissible position set allows for an efficient solution of the uncertainty region recursive updating problem. The proposed recursive approximation algorithms, based on two different classes of approximating regions, exhibit a low computational complexity which is suitable for on-line implementation. Moreover, when the bounded-error assumption is satisfied, a guaranteed outer approximation is provided, i.e., the approximating region always contains the admissible position set.

The paper is structured as follows. Section II introduces the localization problem and the set theoretic approach. Section III provides the main contribution on recursive approximation of the position uncertainty set. A discussion on the computational complexity of the proposed technique and of relevant extensions is provided in Section IV. Section V reports the results of extensive numerical simulation experiments conducted in order to assess the validity of the proposed procedures. Some concluding remarks are given in Section VI.

A. Notation

Let us introduce the notation adopted throughout the paper. Prime denotes transpose; O_n and I_n are the $n \times n$ zero and

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identity matrices, respectively. The i th column of I_n is denoted by e_i . For a vector $v \in \mathbb{R}^n$, $\text{Diag}\{v\}$ denotes the $n \times n$ matrix with $v_i, i = 1, \dots, n$, on the diagonal and zeros elsewhere. The ℓ_∞ and ℓ_1 vector norms are defined in the usual way: $\|v\|_\infty = \max_{i=1, \dots, n} |v_i|$, $\|v\|_1 = \sum_{i=1}^n |v_i|$. The unit ball in the ℓ_∞ norm is denoted by $\mathcal{B}_\infty$. The weighted ℓ_∞ norm of a vector $v \in \mathbb{R}^n$ is defined as $\|v\|_\infty^\omega = \max_{i=1, \dots, n} |v_i/\omega_i|$, where $\omega_i, i = 1, \dots, n$ are positive scalars. The Euclidean norm is denoted by $\|v\|$. Projection matrices from $\mathbb{R}^4$ to $\mathbb{R}^2$ are defined as $\Pi = [I_2 \ 0_2]$ and $\bar{\Pi} = [0_2 \ I_2]$.

Operations between sets are defined in the standard way. The vector sum of two sets $\mathcal{Z}_1$ and $\mathcal{Z}_2$ is given by $\mathcal{Z}_1 + \mathcal{Z}_2 \triangleq \{s = s_1 + s_2, s_1 \in \mathcal{Z}_1, s_2 \in \mathcal{Z}_2\}$, while $M\mathcal{Z} \triangleq \{q : q = Ms, s \in \mathcal{Z}\}$ is the image of a set $\mathcal{Z}$ according to the linear transformation M . The cartesian product of two sets is defined as $\mathcal{Z}_1 \times \mathcal{Z}_2 \triangleq \{s = [s'_1 \ s'_2]', s_1 \in \mathcal{Z}_1, s_2 \in \mathcal{Z}_2\}$. The empty set is denoted by $\emptyset$.

An n -dimensional axis-aligned box is defined as

$$\mathcal{B} = \mathcal{B}(b, c) = \{q \in \mathbb{R}^n : q = c + \text{Diag}\{b\}\alpha, \|\alpha\|_\infty \leq 1\} \quad (1)$$

where c is the center of the box and the absolute values of the elements of b give the size of the edges. An n -dimensional *parallelootope* is defined as

$$\mathcal{P} = \mathcal{P}(T, c) = \{q \in \mathbb{R}^n : q = c + T\alpha, \|\alpha\|_\infty \leq 1\} \quad (2)$$

where c is the center and T is a nonsingular matrix whose column vectors represent the edges of the parallelootope. A parallelootope can be also written as the intersection of n strips, i.e.,

$$\mathcal{P} = \bigcap_{i=1}^n \mathcal{S}(p_i, s_i) \quad (3)$$

where $\mathcal{S}(p_i, s_i) = \{q \in \mathbb{R}^n : |p_i q - s_i| \leq 1\}$ is a *strip* in $\mathbb{R}^n$ (with p_i row vector in $\mathbb{R}^n$) and

$$[p'_1 \dots p'_n]' = T^{-1}, \quad s = T^{-1}c. \quad (4)$$

An m -segment *parpolygon* in $\mathbb{R}^n$ is defined as

$$\mathcal{T}_m = \mathcal{T}(T, c) = \{q \in \mathbb{R}^n : q = c + T\alpha, \|\alpha\|_\infty \leq 1, T \in \mathbb{R}^{n \times m}, \alpha \in \mathbb{R}^m\}. \quad (5)$$

A *parpolygon* is the vector sum of m segments, i.e., $\mathcal{T}_m = c + \sum_{i=1}^m \mathcal{T}(t_i, 0)$, where t_i is the i th column of matrix T . Only bounded *parpolygons* will be considered. This means that it is assumed $m \geq n$ and T full rank. Equivalently, a *parpolygon* is the intersection of $\binom{m}{n-1}$ strips in $\mathbb{R}^n$, which are symmetric with respect to c . Clearly, a parallelootope is a special *parpolygon* with $m = n$.

II. SET MEMBERSHIP LOCALIZATION PROBLEM

Let us consider a vehicle navigating in a two-dimensional (2-D) environment and let $p(t) = [x(t) \ y(t)]'$ be the coordinates of the vehicle location at time t . The vehicle dynamics is described by the linear time-varying model

$$\dot{\xi}(t) = A_c(t)\xi(t) + B_c(t)u(t) + G_c(t)w(t) \quad (6)$$

where $\xi(t) = [x(t) \ y(t) \ \dot{x}(t) \ \dot{y}(t)]'$ is the state vector, containing position and velocity of the vehicle, $u(t) \in \mathbb{R}^2$ is a known input arising from external driving commands and $w(t) \in \mathbb{R}^2$ is the noise term, accounts for unknown forces acting on the vehicle, undesired effects (like wheel slippage or small terrain unevenness) and other unmodeled dynamics. Matrices $A_c(t)$, $B_c(t)$, $G_c(t)$ in (6) may originate from the linearization of a nonlinear model with respect to the current vehicle state (position and velocity). Loosely speaking, these matrices reflect the *a priori* knowledge on the vehicle dynamics (see, e.g., [21]). Since measurements of the environment are collected at finite time instants, it is customary to consider the discretized model

$$\xi(k+1) = A(k)\xi(k) + B(k)u(k) + G(k)w(k) \quad (7)$$

for $k = 0, 1, \dots$, where $\xi(k)$ denotes the state vector at time kT_c and T_c is the sampling time. Matrices $A(k)$, $B(k)$, $G(k)$ are obtained from $A_c(t)$, $B_c(t)$, $G_c(t)$ in (6), in the standard way.

A typical example of dynamic model for a moving vehicle is based on the so-called “low dynamics” assumption (see, e.g., [11]): the vehicle is supposed to move at constant velocity and all the acceleration components are grouped in the unknown term $w(t)$. Therefore, (6) boils down to

$$\dot{\xi}(t) = \begin{bmatrix} 0_2 & I_2 \\ 0_2 & 0_2 \end{bmatrix} \xi(t) + \begin{bmatrix} 0_2 \\ I_2 \end{bmatrix} w(t) \quad (8)$$

and the corresponding discrete-time model is

$$\xi(k+1) = \begin{bmatrix} I_2 & T_c I_2 \\ 0_2 & I_2 \end{bmatrix} \xi(k) + \begin{bmatrix} \frac{T_c^2}{2} I_2 \\ T_c I_2 \end{bmatrix} w(k). \quad (9)$$

The random walk model (9) provides a good approximation of the vehicle dynamics as long as T_c is sufficiently small. A simpler model can be considered if the inputs $u(k)$ represent the x and y displacement measurements provided by odometers, $w(k)$ are the errors of such sensors and velocities are not included in the state vector. In this case, $\xi(k) = p(k) \in \mathbb{R}^2$ and model (7) boils down to $\xi(k+1) = \xi(k) + u(k) + w(k)$.

Due to the presence of uncertainties in the model, the navigating vehicle must collect measurements from the environment to localize itself and to dynamically update its estimated position. It is assumed that a map of the environment is available and n landmarks $l_i, i = 1, \dots, n$ have been identified. Landmark coordinates are known. For each pair of landmarks l_i, l_j , the navigator is able to measure the visual angle, i.e., the angle formed by the rays from the vehicle position to each landmark. The visual angle is a nonlinear function of the current position of the vehicle p and it will be denoted by $\eta_{ij}(p)$. If it were possible to measure the visual angle exactly, the vehicle position would be constrained on a circular arc, whose extreme points are the landmarks l_i, l_j . This is indeed the locus of points from which two (ordered) landmarks are seen under a given visual angle [9]. Unfortunately, in practical situations measurements are not exact and the measured angle is given by

$$\theta_{ij} = \eta_{ij}(p) + v_{ij} \quad (10)$$

where the error term v_{ij} accounts for measurement noise and other uncertainties (for example, errors in landmark positions).

The measurement process can be repeated at different time instants. Exploiting the projection matrix Π , the vehicle position at time k is given by $p(k) = \Pi\xi(k)$ and hence the measurement equation can be written as

$$\theta_{ij}(k) = \eta_{ij}(\Pi\xi(k)) + v_{ij}(k) \quad (11)$$

for $i, j = 1, \dots, n, i < j$ and $k = 0, 1, \dots$. Notice that at each time k , there are in principle $m = n(n-1)/2$ possible visual angle measurements, one for each pair of landmarks. Within the framework outlined above, we introduce the general localization problem.

Localization Problem: Let $\hat{\xi}(0)$ be an estimate of the initial position and velocity of the vehicle. Given the dynamic model (7) and the measurement (11), construct an estimator of the vehicle position $p(k) = \Pi\xi(k)$ at each time instant $k = 1, 2, \dots$

This problem can be tackled in many different ways, depending on the hypotheses on the unknown disturbances $w(k)$ and $v_{ij}(k)$ in (7) and (11). Estimates of the robot position are commonly computed in the literature through the extended Kalman filter [10], [11], [13]. In this paper, a different approach is presented, based on set membership hypotheses on the uncertainties. In particular, it is assumed that the disturbances are unknown-but-bounded, i.e.,

$$|w_i(k)| \leq \varepsilon_i^w(k) \quad i = 1, 2 \quad (12)$$

$$|v_{ij}(k)| \leq \varepsilon_{ij}^v(k) \quad i, j = 1, \dots, n, i < j \quad (13)$$

where $\varepsilon_i^w(k), \varepsilon_{ij}^v(k)$ are known scalars. Equations (12) and (13) can be written in compact form as $\|w(k)\|_\infty^{\varepsilon^w} \leq 1, \|v(k)\|_\infty^{\varepsilon^v} \leq 1$ where $v = [v_{12} \ v_{13} \ \dots \ v_{ij} \ \dots]'$, $\varepsilon^w = [\varepsilon_1^w \ \varepsilon_2^w]'$, $\varepsilon^v = [\varepsilon_{12}^v \ \dots \ \varepsilon_{ij}^v \ \dots]'$. The use of the weighted ℓ_∞ norm gives the possibility of incorporating information on each visual angle measurement in the error bound. Notice that the disturbance bounds are time-varying, to explicitly account for known changes in the environment conditions or in the vehicle dynamics.

Assumption (13) allows one to define a bounded set where the navigator is allowed to lie, for each visual angle measurement θ_{ij} . This set is given by

$$C_{ij} = \mathcal{C}(\theta_{ij}, \varepsilon_{ij}^v) = \{p \in \mathbb{R}^2 : \theta_{ij} - \varepsilon_{ij}^v \leq \eta_{ij}(p) \leq \theta_{ij} + \varepsilon_{ij}^v\} \quad (14)$$

where the dependence on k has been omitted for notational convenience. From a geometrical point of view, C_{ij} is a “thickened ring”, i.e., the region between two circular arcs with the same extreme points, the landmarks l_i, l_j [5], [9] (an example is reported in Fig. 1).

Therefore, at each time k , the visual angle measurements constrain the navigator to the intersection of m thickened rings, which defines the *measurement set*

$$\mathcal{M}(k) = \bigcap_{\substack{i,j=1 \\ i < j}}^n C_{ij}(k). \quad (15)$$

This leads to the formulation of a localization problem in which the dynamic estimate of the position can be expressed in terms of bounded sets.

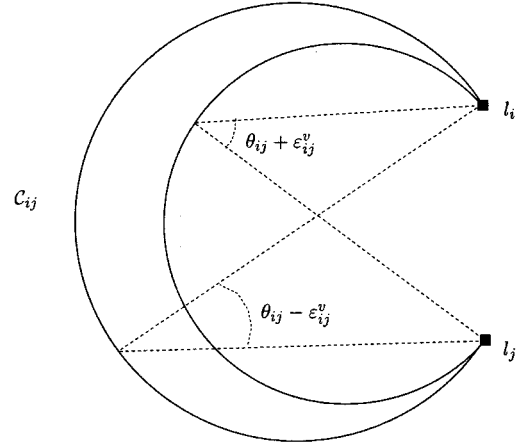


Fig. 1. An example of “thickened ring” C_{ij} .

Set Membership Localization Problem: Let $\Xi(0) \subset \mathbb{R}^4$ be a set containing the initial position and velocity vector $\xi(0)$. Given the dynamic model (7) and the measurement (11), find a set $\mathcal{L}(k)$ containing the vehicle position at time k , i.e., $p(k) \in \mathcal{L}(k)$, $k = 1, 2, \dots$

The solution of the above problem can be obtained from the following recursion, which follows directly from (7) and (11)

$$\Xi(0|0) = \Xi(0) \quad (16)$$

$$\begin{aligned} \Xi(k|k-1) = & A(k-1)\Xi(k-1|k-1) \\ & + B(k-1)u(k-1) \\ & + G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}\mathcal{B}_\infty \end{aligned} \quad (17)$$

$$\Xi(k|k) = \Xi(k|k-1) \cap \{\xi : \Pi\xi \in \mathcal{M}(k)\} \quad (18)$$

$$\mathcal{L}(k) = \Pi\Xi(k|k). \quad (19)$$

The sets $\Xi(k|k-1)$ and $\Xi(k|k)$ are the *feasible state sets*, containing all the positions and velocities that are compatible with the dynamics of the vehicle and the available measurements, up to time $k-1$ and k respectively. Similarly, $\mathcal{L}(k)$ is the *feasible position set*, that includes all possible positions of the navigator at time k (regardless of the velocity). Fig. 2 reports an example of sets $\Xi(k|k-1)$ and $\Xi(k|k)$, for a 2-D model $\xi(k) \in \mathbb{R}^2$ ($\Xi(k-1|k-1)$ is assumed to be a box). Clearly, in this case $\mathcal{L}(k)$ coincides with $\Xi(k|k)$.

Unfortunately, exact computation of the sets $\Xi(k|k-1)$ and $\Xi(k|k)$ in (17)–(18) is a prohibitive task, as the set $\mathcal{M}(k)$ is non-convex and bounded by nonlinear curves. Since we are looking for regions that are guaranteed to contain the vehicle position $p(k)$, outer approximations of the sets $\mathcal{L}(k)$ are sought. In the next section, a recursive approximation strategy exploiting the particular structure of the measurement set $\mathcal{M}(k)$ will be presented.

III. RECURSIVE APPROXIMATIONS OF THE FEASIBLE POSITION SET

In order to reduce the computational burden of the localization algorithm and obtain tractable analytical expressions for the uncertainty regions, simple structure sets containing the exact feasible sets will be considered. In particular, if a simple set

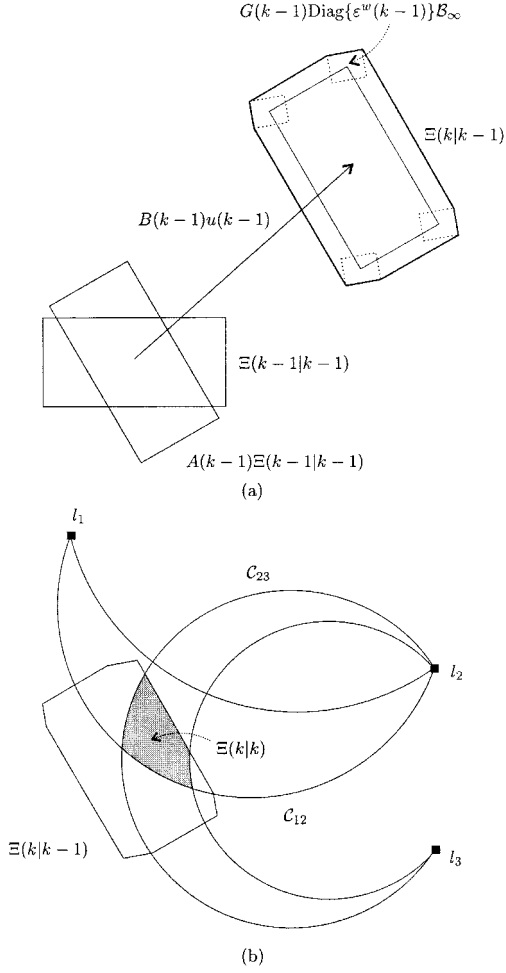


Fig. 2. (a) Time update (17). (b) Measurement update (18).

$\mathcal{R}(k|k-1)$ containing $\Xi(k|k-1)$ is available, one may approximate the feasible position set $\mathcal{L}(k)$ by the intersection of the measurement set $\mathcal{M}(k)$ with $\Pi\mathcal{R}(k|k-1)$. This permits to perform the intersection in (18), which involves the nonconvex set $\mathcal{M}(k)$, in a reduced dimensional space (namely in $\mathbb{R}^2$ instead of $\mathbb{R}^4$). The choice of the approximating sets must be made according to some criterion. Since one is generally interested in reducing as much as possible the size of uncertainty affecting the position and velocity of the navigating vehicle, it seems reasonable to assume that the volume of the sets containing $\Xi(k|k-1)$ and $\Xi(k|k)$ and the area of the set containing $\mathcal{L}(k)$ should be minimized at each time instant k .

Let us now introduce some additional notation. If $\mathcal{R}$ denotes a class of regions of fixed structure, $\overline{\mathcal{R}}(\mathcal{Z})$ is the minimum volume set in the class $\mathcal{R}$, containing the set $\mathcal{Z}$. Let us assume that a set $\mathcal{R}(0|0) \subseteq \mathbb{R}^4$ is given, such that

$$\mathcal{R}(0|0) \supseteq \Xi(0). \quad (20)$$

Then, define the approximating sets $\mathcal{R}(k|k-1)$, $\mathcal{R}(k|k) \subset \mathbb{R}^4$ and $\mathcal{R}_{\Pi}(k|k) \subset \mathbb{R}^2$ through the following recursion

$$\mathcal{R}(k|k-1) = \overline{\mathcal{R}}(A(k)\mathcal{R}(k-1|k-1) + B(k)u(k) + G(k)\text{Diag}\{\varepsilon^w(k)\}\mathcal{B}_\infty) \quad (21)$$

$$\mathcal{R}_{\Pi}(k|k) = \overline{\mathcal{R}}(\Pi\mathcal{R}(k|k-1) \cap \mathcal{M}(k)) \quad (22)$$

$$\mathcal{R}(k|k) = \overline{\mathcal{R}}(\{\mathcal{R}_{\Pi}(k|k) \times \Pi\mathcal{R}(k|k-1)\} \cap \mathcal{R}(k|k-1)). \quad (23)$$

The following proposition states that the set $\mathcal{R}_{\Pi}(k|k)$ is the desired outer approximation of the feasible position set.

Proposition 1: Let $\mathcal{R}_{\Pi}(k|k)$ satisfy the recursion (20)–(23). Then, $p(k) \in \mathcal{L}(k) \subseteq \mathcal{R}_{\Pi}(k|k)$ for all $k = 1, 2, \dots$

Proof: See Appendix

Two recursive algorithms for computing outer approximations of the set $\mathcal{L}(k)$, based on the recursion (20)–(23), will be described in the following. They adopt two different classes of sets as approximating regions $\mathcal{R}$: boxes and parallelotopes [see (1)–(2)].

A. Approximation Through Boxes

In this subsection, the approximating sets $\mathcal{R}$ and $\mathcal{R}_{\Pi}$ will be boxes in $\mathbb{R}^4$ and $\mathbb{R}^2$ respectively. First, notice from (1)–(2) that premultiplication of a box by a square matrix gives a parallelotope. Moreover, for a box $\mathcal{B} \subset \mathbb{R}^4$, $\Pi\mathcal{B}$ is a box in $\mathbb{R}^2$. Hence, according to the recursion (20)–(23), one must solve the following set approximation problems:

- B1) compute the minimum volume box containing the vector sum of two parallelotopes [see (21)];
- B2) compute the minimum area box, containing the intersection of a box in $\mathbb{R}^2$ with the measurement set $\mathcal{M}(k)$ [(22)];
- B3) compute the minimum volume box containing the intersection of a box with the cartesian product of two boxes [(23)]. According to the notation introduced above, $\overline{\mathcal{B}}(\mathcal{Z})$ denotes the minimum volume box containing the set $\mathcal{Z}$. Optimal solutions of Problems B1 and B3 are provided by the following propositions.

Proposition 2: Let $\mathcal{R}(k-1|k-1) = \mathcal{B}(b, c) \subset \mathbb{R}^4$. Then

$$\overline{\mathcal{B}}(A(k-1)\mathcal{B}(b, c) + B(k-1)u(k-1) + G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}\mathcal{B}_\infty) = \mathcal{B}(\bar{b}, \bar{c})$$

where

$$\begin{aligned} \bar{c} &= A(k-1)c + B(k-1)u(k-1) \\ \bar{b}_i &= \|[A(k-1)\text{Diag}\{b\} \\ &\quad G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}]' e_i\|_1 \end{aligned}$$

for $i = 1, \dots, 4$ and e_i denotes the i th column of identity matrix I_4 .

Proof: See Appendix

A 2-D example of Proposition 2 is depicted in Fig. 3.

Proposition 3: Let $\mathcal{R}(k|k-1) = \mathcal{B}(b, c) \subset \mathbb{R}^4$ and $\mathcal{R}_{\Pi}(k|k) = \mathcal{B}(\hat{b}, \hat{c}) \subset \mathbb{R}^2$. Then

$$\begin{aligned} \overline{\mathcal{B}}(\{\mathcal{R}_{\Pi}(k|k) \times \Pi\mathcal{R}(k|k-1)\} \cap \mathcal{R}(k|k-1)) \\ = \mathcal{R}_{\Pi}(k|k) \times \Pi\mathcal{R}(k|k-1) = \mathcal{B}(\bar{b}, \bar{c}) \end{aligned}$$

where

$$\bar{c} = \begin{bmatrix} \hat{c} \\ \Pi\hat{c} \end{bmatrix} \quad \bar{b} = \begin{bmatrix} \hat{b} \\ \Pi\hat{b} \end{bmatrix}.$$

Proof: See Appendix

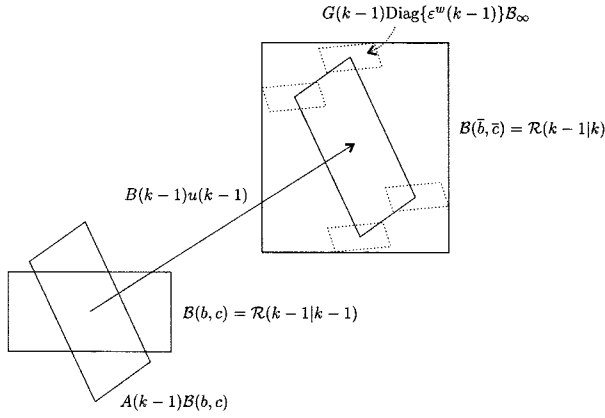


Fig. 3. Box approximation of time update according to Proposition 2 (Problem B1).

Proposition 3: It just states that when boxes are used as approximating region, (23) boils down to the cartesian product of two 2-D boxes in $\mathbb{R}^4$.

Now, let us turn to problem B2. This is the most involved task, since the set $\mathcal{M}(k)$ is the intersection of m thickened rings $\mathcal{C}_{ij}$. In order to obtain a computationally tractable algorithm, a sub-optimal solution based on recursive minimum area approximations is pursued: for each pair i, j , the minimum area box containing the intersection of $\mathcal{C}_{ij}$ with the current approximating box $\mathcal{B}$ is computed. Hence, problem B2 boils down to computing $\bar{\mathcal{B}}(\mathcal{B} \cap \mathcal{C}_{ij})$ for each pair of landmarks l_i, l_j . Let us denote by $\delta\mathcal{C}_{ij}$ the boundary of $\mathcal{C}_{ij}$ in (14). Notice that $\delta\mathcal{C}_{ij} = \delta\mathcal{C}_{ij}^+ \cup \delta\mathcal{C}_{ij}^-$, where $\delta\mathcal{C}_{ij}^+ = \mathcal{C}(\theta_{ij} + \varepsilon_{ij}^v, 0)$ and $\delta\mathcal{C}_{ij}^- = \mathcal{C}(\theta_{ij} - \varepsilon_{ij}^v, 0)$. Moreover, let $\mathcal{V}(\mathcal{B})$ be the set of vertices of the box $\mathcal{B}$. The following result allows one to compute the minimum area box containing $\mathcal{B} \cap \mathcal{C}_{ij}$.

Proposition 4: Let $\mathcal{C}_{ij}$ be given by (14) and $\mathcal{B}$ be an assigned box. Then

$$\bar{\mathcal{B}}(\mathcal{B} \cap \mathcal{C}_{ij}) = \bar{\mathcal{B}}(\mathcal{E})$$

where $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2 \cup \mathcal{E}_3 \cup \mathcal{E}_4$ and

$$\mathcal{E}_1 = \mathcal{V}(\mathcal{B}) \cap \mathcal{C}_{ij} \quad (24)$$

$$\mathcal{E}_2 = \delta\mathcal{B} \cap \delta\mathcal{C}_{ij} \quad (25)$$

$$\mathcal{E}_3 = \{l_i, l_j\} \cap \mathcal{B} \quad (26)$$

$$\mathcal{E}_4 = \delta(\bar{\mathcal{B}}(\delta\mathcal{C}_{ij}^+)) \cap \delta\mathcal{C}_{ij}^- \cap \mathcal{B}. \quad (27)$$

Proof: See Appendix

We remark that the sets $\mathcal{E}_i, i = 1, \dots, 4$ in Proposition 4 contain a finite number of points, which are very easy to compute. For example, the points in $\mathcal{E}_2$ are given by the intersection of a circle and a segment, while $\mathcal{E}_4$ requires the computation of the tangency point between a line and a circle. An example in which all the four sets $\mathcal{E}_i$ are not empty is reported in Fig. 4.

The important implication of Proposition 4 is that the finite set of points $\mathcal{E}$ is the only thing one needs to know in order to compute the minimum area box containing $\mathcal{B} \cap \mathcal{C}_{ij}$. Therefore, if $\Pi\mathcal{R}(k|k-1) = \mathcal{B}(b, c) \subset \mathbb{R}^2$, $\mathcal{C}_{ij}(k)$ is given by (14) and $\mathcal{E}$ is defined as in Proposition 4, then

$$\bar{\mathcal{B}}(\Pi\mathcal{R}(k|k-1) \cap \mathcal{C}_{ij}(k)) = \mathcal{B}(\bar{b}, \bar{c}) \quad (28)$$

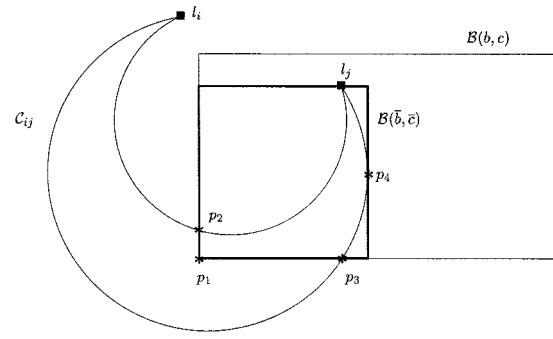


Fig. 4. Box approximation of the intersection between a thickened ring $\mathcal{C}_{ij}$ and a given box $\mathcal{B}(b, c)$ (Problem B2). The four sets in Proposition 4 are, respectively: $\mathcal{E}_1 = \{p_1\}$, $\mathcal{E}_2 = \{p_2, p_3\}$, $\mathcal{E}_3 = \{p_4\}$, $\mathcal{E}_4 = \{l_j\}$. The resulting approximating box $\mathcal{B}(\bar{b}, \bar{c})$ is the thick one.

where

$$\begin{aligned} \bar{c}_1 &= \frac{\bar{x} + \underline{x}}{2}, & \bar{c}_2 &= \frac{\bar{y} + \underline{y}}{2} \\ \bar{b}_1 &= \frac{\bar{x} - \underline{x}}{2}, & \bar{b}_2 &= \frac{\bar{y} - \underline{y}}{2} \end{aligned} \quad (29)$$

and $\bar{x} = \max_{p \in \mathcal{E}} x$, $\underline{x} = \min_{p \in \mathcal{E}} x$, $\bar{y} = \max_{p \in \mathcal{E}} y$, $\underline{y} = \min_{p \in \mathcal{E}} y$.

According to the recursive strategy proposed for the sub-optimal solution of problem B2, the set $\mathcal{R}_{\Pi}(k|k)$ can be computed in the following way:

- set $\mathcal{B} = \Pi\mathcal{R}(k|k-1)$;
- for $i, j = 1, \dots, n, i < j$, set $\mathcal{B} = \bar{\mathcal{B}}(\mathcal{B} \cap \mathcal{C}_{ij})$ using (28)–(29);
- set $\mathcal{R}_{\Pi}(k|k) = \mathcal{B}$.

B. Approximation Through Parallelotopes

A more accurate and still viable alternative to box approximations, is to choose parallelotopes in $\mathbb{R}^4$ and $\mathbb{R}^2$ as approximating sets $\mathcal{R}$ and $\mathcal{R}_{\Pi}$. The main advantage of using parallelotopes instead of boxes is that the orientation of the approximating region can be changed: this allows one to reduce the size of the uncertainty set, especially when the true feasible set stretches along some direction in the state space (or in the position plane). The overall strategy is the same as in Section III-A. The criterion for the selection of the approximating parallelotopes is still minimum volume (or area).

From (2) it is easy to see that premultiplication of a parallelotope by a square matrix gives a parallelotope. Moreover, for a parallelotope $\mathcal{P} \subset \mathbb{R}^4$, $\Pi\mathcal{P}$ is a 4-segment *parpolygon* in $\mathbb{R}^2$. Hence, the approximation problems arising from recursion (20)–(23) with parallelotopes as sets $\mathcal{R}$, are as follows:

- P1) compute the minimum volume parallelotope containing the vector sum of two parallelotopes [see (21)];
- P2) compute the minimum area parallelotope, containing the intersection of a parpolygon in $\mathbb{R}^2$ with $\mathcal{M}(k)$ [(22)];
- P3) compute the minimum volume parallelotope containing the intersection of a parallelotope with the cartesian product of a parallelotope and a parpolygon [(23)].

Approximate solutions of problems P1 and P3 can be obtained by exploiting the techniques proposed in [22], [23]. In particular, the following results are needed.

Theorem 1: Consider the parpolygon

$$T = c + \sum_{i=1}^{n+1} T(t_i, 0), \quad t_i \in \mathbb{R}^n$$

and let the matrix $[t_1, \dots, t_n]$ be nonsingular, with $[p'_1 \dots p'_n]' = [t_1, \dots, t_n]^{-1}$. Then the parallelotope of minimum volume outbounding T is given by

$$\bar{P}(T) = c + \sum_{\substack{i=1 \\ i \neq i^*}}^{n+1} T(\bar{t}_i, 0_n) = \mathcal{P}(T^*, c)$$

where T^* is obtained by replacing the i^* th column of $[t_1, \dots, t_n]$ by t_{n+1} and

$$\begin{aligned} i^* &= \arg \max_{1 \leq i \leq n+1} |p_i t_{n+1}| \text{ with } p_{n+1} t_{n+1} \triangleq 1 \\ \bar{t}_i &= r_i t_i, \quad 1 \leq i \leq n+1, \quad i \neq i^* \\ r_i &= 1 + \left(\frac{|p_i t_{n+1}|}{|p_{i^*} t_{n+1}|} \right), \quad 1 \leq i \leq n+1, \quad i \neq i^*. \end{aligned}$$

Theorem 2: Consider the convex polytope

$$\mathcal{X} = \bigcap_{i=1}^{n+1} \mathcal{S}(p_i, s_i)$$

and let the matrix $[p'_1, \dots, p'_n]'$ be nonsingular, with $[t_1, \dots, t_n] = ([p'_1 \dots p'_n]')^{-1}$. Then, the parallelotope of minimum volume outbounding $\mathcal{X}$ is given by

$$\bar{P}(\mathcal{X}) = \bigcap_{\substack{i=1 \\ i \neq i^*}}^{n+1} \mathcal{S}(\bar{p}_i, \bar{s}_i) = \mathcal{P}(T^*, c^*)$$

where $T^* = (P^*)^{-1}$, with P^* obtained by replacing the i^* th row of $[p'_1 \dots p'_n]'$ by $\bar{p}_{n+1}$, $c^* = T^* \bar{s}$ and

$$\begin{aligned} i^* &= \arg \max_{1 \leq i \leq n+1} |p_{n+1} t_i| \text{ with } p_{n+1} t_{n+1} \triangleq 1 \\ \bar{p}_i &= \frac{2}{r_i^+ - r_i^-} p_i \\ \bar{s}_i &= \frac{2}{r_i^+ - r_i^-} \left(s_i + \frac{r_i^+ + r_i^-}{2} \text{sign}(p_{n+1} t_i) \right) \\ r_i^+ &= \begin{cases} \min \left(-1 + \frac{1 - \epsilon_{n+1}^-}{|p_{n+1} t_i|}, 1 \right) & \text{if } p_{n+1} t_i \neq 0 \\ 1 & \text{if } p_{n+1} t_i = 0 \end{cases} \\ r_i^- &= \begin{cases} \max \left(1 - \frac{1 + \epsilon_{n+1}^-}{|p_{n+1} t_i|}, -1 \right) & \text{if } p_{n+1} t_i \neq 0 \\ -1 & \text{if } p_{n+1} t_i = 0 \end{cases} \\ &\text{for } i = 1, \dots, n \\ r_{n+1}^+ &= \min(\epsilon_{n+1}^+, 1) \\ r_{n+1}^- &= \max(\epsilon_{n+1}^-, -1) \\ \epsilon_{n+1}^+ &= (p_{n+1} \hat{x} - s_{n+1}) + \sum_{i=1}^n |p_{n+1} t_i| \\ \epsilon_{n+1}^- &= (p_{n+1} \hat{x} - s_{n+1}) - \sum_{i=1}^n |p_{n+1} t_i|. \end{aligned}$$

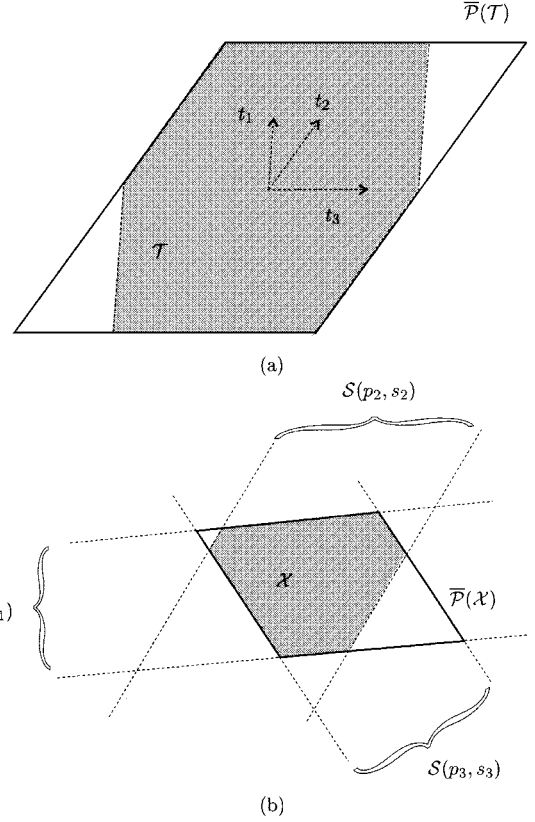


Fig. 5. (a) Example of Theorem 1, for recursive approximation of problem P1. (b) Example of Theorem 2, for recursive approximation of problem P3.

Examples of the approximations provided by Theorems 1 and 2 are presented in Fig. 5, for the case $n = 2$.

Now, we show how these results can be exploited to compute recursively approximate solutions of problems P1 and P3. Let us consider problem P1 and assume $\mathcal{R}(k-1|k-1) = \mathcal{P}(T, c) \subset \mathbb{R}^4$. According to (5), (21) can be rewritten as

$$\mathcal{R}(k|k-1) = \bar{P}(\mathcal{T}_8) \quad (30)$$

where

$$\begin{aligned} \mathcal{T}_8 &= T([A(k-1)T|G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}], \\ &\quad A(k-1)c + B(k-1)u(k-1)), \end{aligned}$$

and as usual $\bar{P}(\mathcal{Z})$ denotes the minimum volume parallelotope containing $\mathcal{Z}$. A suboptimal solution of problem (30) can be computed by recursively applying Theorem 1 to $\mathcal{T}_8$. The idea is as follows:

- let $\bar{c} = A(k-1)c + B(k-1)u(k-1)$ and set $\bar{T}^{(0)} = A(k-1)T$, $[t_1 \ t_2 \ t_3 \ t_4] = G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}$;
- for $i = 1, \dots, 4$, compute $\mathcal{P}(\bar{T}^{(i)}, \bar{c}) = \bar{P}\left(T([\bar{T}^{(i-1)} \mid t_i], \bar{c})\right)$ using Theorem 1 ;
- set $\mathcal{R}(k|k-1) = \mathcal{P}(\bar{T}^{(4)}, \bar{c})$.

Clearly, the above construction guarantees that

$$\begin{aligned} \mathcal{R}(k|k-1) &= \mathcal{P}(\bar{T}^{(4)}, \bar{c}) \supseteq A(k-1)\mathcal{R}(k-1|k-1) \\ &\quad + B(k-1)u(k-1) + G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}\mathcal{B}_\infty. \end{aligned}$$

Turning to problem P3, let us assume $\mathcal{R}(k|k-1) = \mathcal{P}(T, c) \subset \mathbb{R}^4$ and $\mathcal{R}_\Pi(k|k) = \mathcal{P}(\hat{T}, \hat{c}) \subset \mathbb{R}^2$. Then, (23) boils down to

$$\mathcal{R}(k|k) = \overline{\mathcal{P}}(\mathcal{T}_6 \cap \mathcal{P}(T, c)) \quad (31)$$

where

$$\mathcal{T}_6 = \mathcal{T} \left(\begin{bmatrix} \hat{T} & 0 \\ 0 & \Pi T \end{bmatrix}, \begin{bmatrix} \hat{c} \\ \Pi c \end{bmatrix} \right).$$

A suboptimal solution of problem (31) can be computed in two steps:

- 1) Apply Theorem 1 to $\mathcal{T}_6$, as it has been done above for $\mathcal{T}_8$ in problem P1, thus obtaining $\mathcal{P}(\tilde{T}, \tilde{c}) \supset \mathcal{T}_6$.
- 2) Recursively apply Theorem 2 to the intersection between $\mathcal{P}(\tilde{T}, \tilde{c})$ and $\mathcal{P}(T, c)$. This can be done in the following way:
 - set $\tilde{T}^{(0)} = \tilde{T}$, $\tilde{c}^{(0)} = \tilde{c}$ and $[p'_1 \dots p'_4]' = T^{-1}$, $s = T^{-1}c$;
 - for $i = 1, \dots, 4$, use Theorem 2 to compute $\mathcal{P}(\tilde{T}^{(i)}, \tilde{c}^{(i)}) = \overline{\mathcal{P}}(\mathcal{P}(\tilde{T}^{(i-1)}, \tilde{c}^{(i-1)}) \cap \mathcal{S}(p_i, s_i))$
 - let $\mathcal{R}(k|k) = \mathcal{P}(\tilde{T}^{(4)}, \tilde{c}^{(4)})$.

Once again, the above procedure provides a guaranteed approximation of the feasible set, i.e.,

$$\mathcal{R}(k|k) = \mathcal{P}(\tilde{T}^{(4)}, \tilde{c}^{(4)}) \supseteq \{\mathcal{R}_\Pi(k|k) \times \Pi \mathcal{R}(k|k-1)\} \cap \mathcal{R}(k|k-1).$$

A suboptimal solution of Problem P2 can be obtained adopting a recursive strategy similar to that proposed for boxes in Section III-A. Recall from (22) that one has to find a parallelotope containing the intersection of $\mathcal{M}(k)$ with the parpolygon $\Pi \mathcal{R}(k|k-1)$, where $\mathcal{R}(k|k-1)$ is given by (30). One can first exploit the same technique adopted to solve problem P1 (based on Theorem 1), to find a parallelotope containing $\Pi \mathcal{R}(k|k-1)$. Then, the main idea is that for each pair i, j , the minimum area parallelotope containing the intersection of the thickened ring $\mathcal{C}_{ij}$ with the current approximating parallelotope $\mathcal{P}$ must be computed. Because parallelotopes have more degrees of freedom than boxes, the intersection with $\mathcal{C}_{ij}$ can not only shrink the approximating parallelotope, but also change its shape (i.e., the orientation of the strips). We consider shrinking and reshaping separately.

First, let us denote by $\overline{\mathcal{P}}_T(\mathcal{Z})$ the minimum parallelotope of fixed shape T containing the set $\mathcal{Z}$, i.e., $\overline{\mathcal{P}}_T(\mathcal{Z}) = \mathcal{P}(TD^*, c^*)$ where

$$\{D^*, c^*\} = \arg \min_{s.t.} \det(TD) \quad \{D \text{ diagonal}, c \in \mathbb{R}^2 : \mathcal{P}(TD, c) \supseteq \mathcal{Z}\}. \quad (32)$$

Then, we have the following result, which is a straightforward extension of Proposition 4.

Proposition 5: Let $\mathcal{C}_{ij}$ be given by (14) and $\mathcal{P} = \mathcal{P}(T, c)$ be an assigned parallelotope. Then

$$\overline{\mathcal{P}}_T(\mathcal{P} \cap \mathcal{C}_{ij}) = \overline{\mathcal{P}}(\mathcal{F})$$

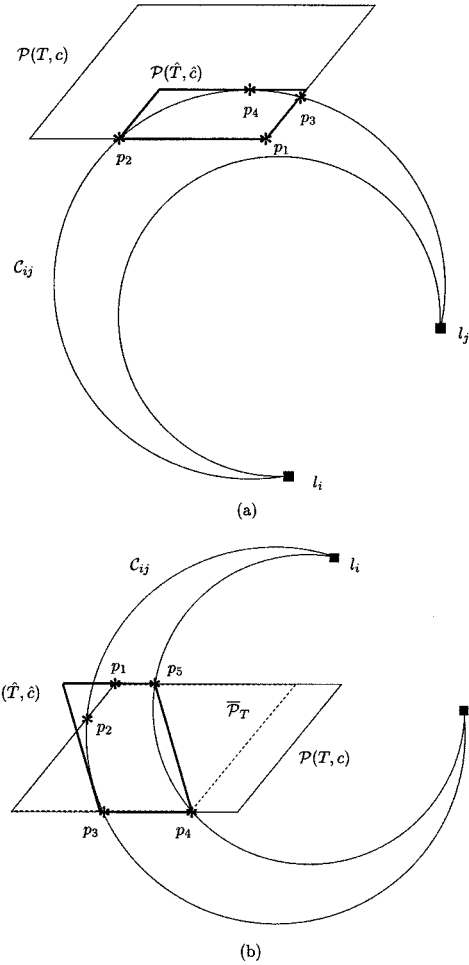


Fig. 6. Parallelotopic approximation of the intersection between a thickened ring $\mathcal{C}_{ij}$ and a given parallelotope $\mathcal{P}(T, c)$ (Problem P2). (a) A case in which the shape of the parallelotope does not change: $\mathcal{F}_1 = \{p_1\}$, $\mathcal{F}_2 = \{p_2, p_3\}$, $\mathcal{F}_3 = \emptyset$, $\mathcal{F}_4 = \{p_4\}$; the resulting tightened parallelotope $\mathcal{P}(\hat{T}, \hat{c})$ is the thick one. Notice that $\mathcal{F}_s = \emptyset$. (b) A case in which the shape of the parallelotope changes: $\mathcal{F}_1 = \{p_1\}$, $\mathcal{F}_2 = \{p_2, p_3, p_4, p_5\}$, $\mathcal{F}_3 = \mathcal{F}_4 = \emptyset$ and $\mathcal{F}_s = \{p_4, p_5\}$; the tightened parallelotope $\overline{\mathcal{P}}_T$ is dashed, the resulting one $\mathcal{P}(\hat{T}, \hat{c})$ is thick.

where $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2 \cup \mathcal{F}_3 \cup \mathcal{F}_4$ and

$$\mathcal{F}_1 = \mathcal{V}(\mathcal{P}) \cap \mathcal{C}_{ij}, \quad (33)$$

$$\mathcal{F}_2 = \delta \mathcal{P} \cap \delta \mathcal{C}_{ij}, \quad (34)$$

$$\mathcal{F}_3 = \{l_i, l_j\} \cap \mathcal{P}, \quad (35)$$

$$\mathcal{F}_4 = \delta(\overline{\mathcal{P}}_T(\delta \mathcal{C}_{ij}^-)) \cap \delta \mathcal{C}_{ij}^-. \quad (36)$$

As for Proposition 4, the sets $\mathcal{F}_i$, $i = 1, \dots, 4$, contain a finite number of points that are very easy to compute (see Fig. 6). Using Proposition 5, it is possible to give an explicit characterization of the set $\overline{\mathcal{P}}_T(\mathcal{P} \cap \mathcal{C}_{ij})$. Let a parallelotope $\mathcal{P} = \mathcal{P}(T, c) \subset \mathbb{R}^2$ and a thickened ring $\mathcal{C}_{ij}$ be given and let $[p'_1 \ p'_2]' = T^{-1}$. Moreover, let $\mathcal{F}$ be defined as in Proposition 5. Then,

$$\overline{\mathcal{P}}_T(\mathcal{P} \cap \mathcal{C}_{ij}) = \bigcap_{i=1}^2 \mathcal{S}(\overline{p}_i, \overline{s}_i) \quad (37)$$

where

$$\begin{aligned} \bar{p}_i &= \frac{2}{\gamma_i^+ - \gamma_i^-} p_i, \quad \bar{s}_i = \frac{\gamma_i^+ + \gamma_i^-}{\gamma_i^+ - \gamma_i^-}, \\ \gamma_i^+ &= \max_{q \in \mathcal{F}} p_i q, \quad \gamma_i^- = \max_{q \in \mathcal{F}} p_i q. \end{aligned} \quad (38)$$

Notice that the optimal solution D^*, c^* of problem (32) can be easily obtained from $\bar{p}_i, \bar{s}_i, i = 1, 2$ in (38), using parallelotope definitions (2)–(3) and (4).

Now, let $\bar{\mathcal{P}}_T$ denote the tightened parallelotope $\bar{\mathcal{P}}_T(\mathcal{P} \cap \mathcal{C}_{ij})$ and let us try to further reduce the volume of $\bar{\mathcal{P}}_T$ by changing its shape, i.e., by substituting new strips to the $\mathcal{S}(\bar{p}_i, \bar{s}_i), i = 1, 2$, in (37)–(38). Let us consider the set

$$\mathcal{F}_s = \mathcal{F}_3 \cup \{\delta\mathcal{C}_{ij}^+ \cap \bar{\mathcal{P}}_T\}. \quad (39)$$

Recall that $\mathcal{F}_3$ is the set of landmarks lying inside the initial parallelotope $\mathcal{P}$ (and hence also inside $\bar{\mathcal{P}}_T$), while $\delta\mathcal{C}_{ij}^+$ is the inner circle bounding the thickened ring $\mathcal{C}_{ij}$. It is clear that $\mathcal{F}_s$ is made of a finite set of points; more precisely, it can contain from zero up to eight points (the maximum number of intersections between a circle and a parallelotope in $\mathbb{R}^2$). However, the most common situations are as follows: i) $\mathcal{F}_s$ is empty; ii) $\mathcal{F}_s$ contains two points. When $\mathcal{F}_s$ is empty it is easy to check that reshaping cannot reduce the size of the approximating parallelotope, i.e.,

$$\bar{\mathcal{P}}(\mathcal{P}(T, c) \cap \mathcal{C}_{ij}) = \bar{\mathcal{P}}_T.$$

Let us consider case ii) and set $\mathcal{F}_s = \{q_1, q_2\}$. Let v be a row vector orthogonal to $q_1 - q_2$ (i.e., satisfying $v(q_1 - q_2) = 0$) and denote by $\bar{\mathcal{S}}_v(\mathcal{Z})$ the minimum width strip orthogonal to v containing the set $\mathcal{Z}$. From the definition of strip, this is given by $\mathcal{S}(\alpha^* v, s^*)$, where α^* is picked as the maximum positive α such that $\mathcal{S}(\alpha v, s^*) \supset \mathcal{C}_{ij}$ for some $s^* \in \mathbb{R}$. The computation of $\mathcal{S}(\alpha^* v, s^*)$ simply requires the computation of the tangency point between a line of direction $q_1 - q_2$ and the circle $\delta\mathcal{C}_{ij}^-$ (the outer circle bounding $\mathcal{C}_{ij}$). The strip $\mathcal{S}(\alpha^* v, s^*)$ is used as a candidate strip for replacing one of the two strips of $\bar{\mathcal{P}}_T$. The minimum volume parallelotope $\mathcal{P}(\hat{T}, \hat{c})$ containing the intersection of the three strips $\mathcal{S}(\bar{p}_1, \bar{s}_1), \mathcal{S}(\bar{p}_2, \bar{s}_2)$ and $\mathcal{S}(\alpha^* v, s^*)$, can be easily obtained by applying Theorem 2. Notice that by construction we have

$$\begin{aligned} \mathcal{P}(\hat{T}, \hat{c}) &= \bar{\mathcal{P}}(\mathcal{S}(\alpha^* v, s^*) \cap \mathcal{S}(\bar{p}_1, \bar{s}_1) \cap \mathcal{S}(\bar{p}_2, \bar{s}_2)) \\ &\supset \{\mathcal{P}(T, c) \cap \mathcal{C}_{ij}\}. \end{aligned} \quad (40)$$

Fig. 6 shows two examples of the parallelotopic approximation of $\mathcal{P}(T, c) \cap \mathcal{C}_{ij}$, computed according to the above procedure. In Fig. 6(a), the set $\mathcal{F}_s$ is empty because there are no landmarks inside the initial parallelotope $\mathcal{P}(T, c)$ ($\mathcal{F}_3 = \emptyset$) and the inner bound of the thickened ring does not intersect the tightened parallelotope $\bar{\mathcal{P}}_T$ (the thick one in the figure). Hence, the initial parallelotope is only tightened according to Proposition 5, but its shape does not change. On the contrary, Fig. 6(b) shows a case in which $\mathcal{F}_s$ is not empty, as $\delta\mathcal{C}_{ij}^+$ cuts the tightened parallelotope $\bar{\mathcal{P}}_T$ (dashed) in the points p_4, p_5 . Notice that the resulting parallelotope $\mathcal{P}(\hat{T}, \hat{c})$ (thick) is much smaller than the tightened one.

Finally, when $\mathcal{F}_s$ contains more than two points, the intersection $\mathcal{P} \cap \mathcal{C}_{ij}$ is made up of two or more disjoint subsets. A pos-

sible strategy may be to approximate each subset by a new parallelotope. A conservative but less computationally demanding alternative is to keep the tightened parallelotope $\bar{\mathcal{P}}_T(\mathcal{P} \cap \mathcal{C}_{ij})$ as approximating parallelotope containing $\mathcal{P} \cap \mathcal{C}_{ij}$, without selecting any new strip. This does not cause any problem in most practical situations, as the ambiguity between disjoint feasible subsets are usually resolved by the intersection with thickened rings associated to other measurements.

Summing up, the following recursive procedure can be employed for the computation of the set $\mathcal{R}_{\Pi}(k|k)$ in problem P2:

- compute the parpolygon $\mathcal{I}_4 = \Pi\mathcal{R}(k|k-1)$;
- set $\mathcal{P} = \bar{\mathcal{P}}(\mathcal{I}_4)$ (using the same technique, based on Theorem 1, adopted for computing (30));
- for $i, j = 1, \dots, n, i < j$, set

$$\mathcal{P} = \begin{cases} \mathcal{P}(\hat{T}, \hat{c}) \text{ in (40),} & \text{if } \mathcal{F}_s \text{ contains two points,} \\ \bar{\mathcal{P}}_T(\mathcal{P} \cap \mathcal{C}_{ij}), & \text{otherwise;} \end{cases}$$

- set $\mathcal{R}_{\Pi}(k|k) = \mathcal{P}$.

IV. DISCUSSION OF SET APPROXIMATION ALGORITHMS

In this section, the computational complexity of the set membership localization technique is analyzed. Moreover, some relevant extensions of the proposed approach are briefly discussed.

A. Computational Complexity

The set membership localization algorithms presented in Section III enjoy a low computational complexity, thanks to the fact that the true feasible sets are recursively approximated via simple sets, like boxes or parallelotopes. Consider for example the box-based approximation procedure described in Section III-A. Tasks B1 and B3 can be accomplished according to Propositions 2 and 3 and require very few operations (products of matrices and vectors in a space of dimension 4). The most complex task is B2, whose suboptimal recursive solution is based on the computation of the smallest box containing the intersection of a box and a thickened ring, for each landmark pair. This can be obtained via Proposition 4, which requires the computation of a small number of points, as intersections of segments and circles in $\mathbb{R}^2$. Hence, the overall computational burden of recursion (21)–(23) at time k basically depends on the number of considered landmark pairs. For n landmarks, this can be at most $n(n-1)/2$ and therefore the computational complexity of the localization landmark turns out to be $O(n^2)$. If the robot is equipped with a panoramic vision system [24] or a rotating laser scanner [2], performing one 360° circular scan of the environment and collecting measurements of visual angles between pairs of consecutive landmarks, one has n thickened rings and the complexity reduces to $O(n)$.

Analogous observations apply to the parallelotope-based algorithm of Section III-B, whose only additional computations required are those relative to the set $\mathcal{F}_s$ in (39), which again contains points given by intersections of segments and circles in $\mathbb{R}^2$.

In general, the computational burden of the set membership localization algorithm depends on the number of landmarks detected at time k . If the robot is able to scan only a limited part of the environment (for example, due to use of a camera with

restricted vision field) the average complexity will be $O(m)$, where m is the average number of visual angle measurements processed at each time instant. Clearly, if very few landmarks are detected, the localization will be more rough, resulting in larger feasible sets.

It is also worth observing that due to the approximations involved in the recursive procedure, reprocessing of measurements may be useful to further reduce the size of the approximate feasible sets, as it is well known in set membership estimation literature (see, e.g., [25]). In this respect, an objective to be pursued is to suitably trade-off the increase of the computational burden due to data reprocessing and the uncertainty reduction in the approximation of feasible sets.

B. Extensions

The proposed set membership technique can be extended to more complex scenarios, without affecting the basic approach and with a limited increase of the computational burden.

The assumption that landmarks can be modeled as points in $\mathbb{R}^2$ can be easily relaxed. Indeed, non negligible landmark size can be treated by enlarging the bounds ε_{ij}^v on visual angle measurement errors. In principle, this can be done also to account for small uncertainties in the position of landmarks, which can be supposed to be bounded by a fixed quantity.

A much more challenging problem arises when the position of landmarks is unknown and the robot has to both localize the landmarks and estimate its own position with respect to them. This is the so-called *simultaneous localization and map building problem*, which has been recently addressed in the literature in the classical probabilistic framework [3], [26] and also via a set membership approach [27]. The greatest difficulty when facing this problem originates from the fact that the state vector to be estimated includes all landmark positions and therefore its dimension in no less than $2(n+1)$ no matter which motion model is adopted.

In this paper, only measurements of visual angles between a pair of landmarks are considered, according to what is done in many papers in the literature (see, e.g., [4]–[8]). However, the proposed set membership approach is able to cope with different type of measurements, like, e.g.,

- *distance* measurements between the robot and a landmark, provided by a laser rangefinder or a stereo couple;
- measurements of *absolute orientation*, i.e., of the angle between the direction from the robot to a landmark and a fixed direction; the latter can be defined by the current robot heading or by a fixed direction detected by absolute orientation sensors (like a compass or a solar sensor).

Assuming bounded additive errors for the above measurements, one can define the corresponding measurement set, similarly to what has been done in (14)–(15). Specifically, each distance measurement defines a feasible circular corona in $\mathbb{R}^2$, while an absolute orientation measurement gives a feasible conic sector. The recursive set approximation techniques in Section III can be easily extended to these measurements sets.

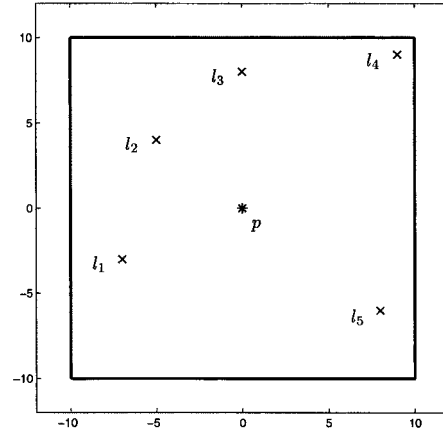


Fig. 7. The simulation environment (thick box: the room; * : robot position; x : landmarks).

Another relevant extension concerns the estimation of robot orientation. This can be included in the state vector, whose dimension turns out to be 6 (reducing to 3 for the simplified motion model that does not consider velocities). Simultaneous estimation of position and orientation leads to feasible state sets of higher dimension, which slightly complicates the set approximation procedure. A viable alternative may consist of estimating 2-D feasible position sets and 1-D feasible orientation interval. The correlation between position and orientation can be exploited by iteratively refining the position set with respect to the current admissible orientations and vice versa (see [28] for details).

V. NUMERICAL SIMULATION EXPERIMENTS

In this section, simulation results of some localization experiments are reported. The set membership localization algorithms providing box and parallelotopic approximations of the feasible position set have been tested in both static and dynamic setting.

A. Static Setting

First, we consider a situation in which the robot is still and has to locate itself with respect to the landmarks. At a fixed time k , the navigating vehicle is supposed to be located at the center of a square room, of 20 meters side. This means that $\Pi\mathcal{R}(k|k-1) = \Pi\Xi(k|k-1) = \mathcal{B}([10 \ 10]', [0 \ 0]')$, in a reference system centered at the vehicle position (see Fig. 7). The sets $\mathcal{R}_{\Pi}(k|k)$ defined in (22) are computed according to the recursive strategies for the solutions of problems B2 and P2, outlined in Section III-A, B. The centers of boxes and parallelotopes have been considered as nominal estimates $\hat{p}$ of the vehicle position.

In a first set of experiments, it is assumed that five landmarks are identified in the scene. The visual angle measurements are corrupted by additive noise $v_{ij}(k)$, generated as an independent uniformly distributed signal satisfying (13) with constant bound $\varepsilon_{ij}^v(k) = \varepsilon^v, \forall i, j$. Error bounds for commercial sensors range from $0.5^\circ \div 1^\circ$ of popular rotating laser scanners, to much smaller errors if sophisticated panoramic vision systems are employed. Nevertheless, also larger error bounds have been considered in the simulations, in order to account

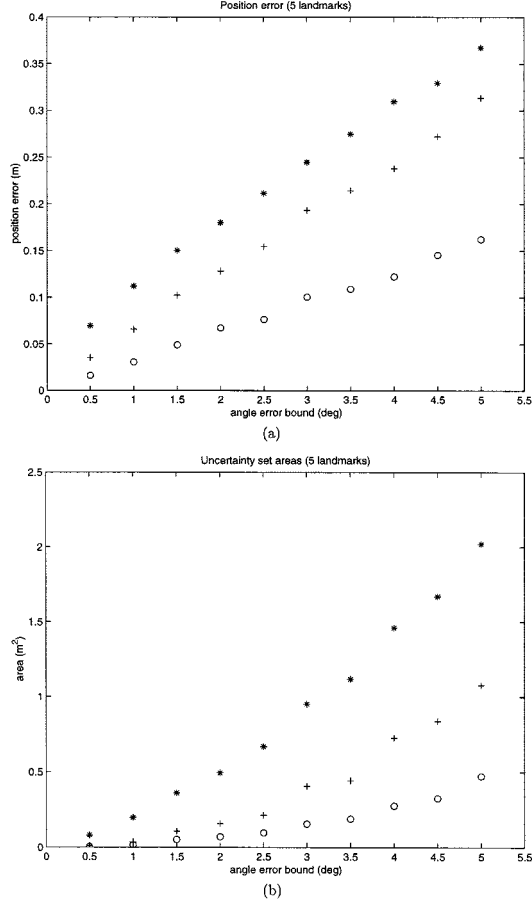


Fig. 8. Localization with five landmarks. (a) Average nominal position errors. (b) Average uncertainty set areas for different visual angle noise bounds (*: boxes; +: parallelograms; o: minimum box containing the exact feasible position set $\mathcal{L}(k)$).

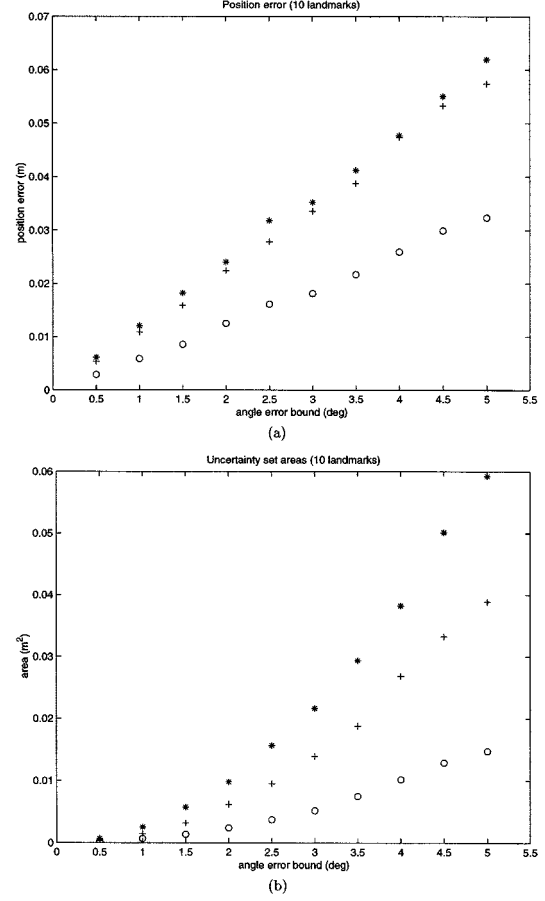


Fig. 9. Localization with 10 landmarks. (a) Average nominal position errors. (b) Average uncertainty set areas for different visual angle noise bounds (*: boxes; +: parallelograms; o: minimum box containing the exact feasible position set $\mathcal{L}(k)$).

for non negligible landmark size and errors in landmark positions, as explained in Section IV-B. The nominal localization errors $\|\hat{p} - p\|$ (in meters), obtained for different values of ε^v ranging from 0.5 to 5 degrees, are reported in Fig. 8(a). Results are averaged over 1000 different landmark configurations in the square room. In Fig. 8(b), the area of the approximating boxes and parallelograms are compared with that of the minimum area box containing the exact feasible position set $\mathcal{L}(k)$, i.e., $\bar{\mathcal{B}}(\mathcal{L}(k)) = \bar{\mathcal{B}}(\Pi\Xi(k|k-1) \cap \mathcal{M}(k))$. Notice that the parallelogramic uncertainty sets are smaller than the corresponding boxes and that their area is quite close to that of the minimum outer box $\bar{\mathcal{B}}(\mathcal{L}(k))$ (whose computation has been performed by gridding a sufficiently large area containing the true robot position: notice that this has required a much higher computational burden, compared to the proposed approximation algorithms). The same experiment is repeated for the case of 10 landmarks, in the above setting. Results concerning nominal position errors and uncertainty set areas are reported in Fig. 9. It can be noticed that nominal position errors and areas of the uncertainty sets have been remarkably reduced.

Another set of experiments is performed, assuming a relative visual angle error bound. In these simulations, ε_{ij}^v is set to 20% of the current visual angle measure θ_{ij} . The estimated nominal positions for 100 different settings including five landmarks are

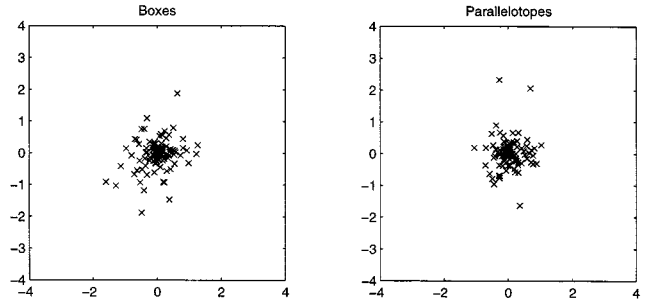


Fig. 10. Position estimates for 100 different groups of five landmarks: $\varepsilon_{ij}^v = 0.2\theta_{ij}$, $p = [0 \ 0]'$.

reported in Fig. 10. It can be observed that, despite the large measurement error, the “cloud” of nominal estimates is not too scattered. The average nominal position error is 0.49m for parallelograms and 0.51m for boxes. Average uncertainty areas are respectively 2.32 m² and 5.42 m². Once again, nominal errors provided by the two algorithms are similar, but the parallelogramic approximation gives smaller uncertainty sets.

The same experiment is repeated placing the vehicle almost at one corner of the square room. In this case, the visual angles range only from 0 to 90 degrees, thus reducing the maximum size of measurement noise. The results are depicted in Fig. 11,

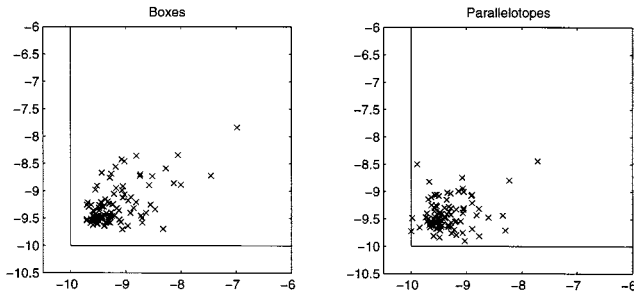


Fig. 11. Position estimates for 100 different groups of five landmarks: $\varepsilon_{ij}^v = 0.2\theta_{ij}$, $p = [-9.5 - 9.5]'$.

where average nominal position errors are 0.45 m for parallelotopes and 0.51 m for boxes. Average uncertainty areas are reduced to 1.77 m² and 2.07 m², respectively. The results obtained in all the performed experiments appear of good quality, when compared to those presented in recent literature addressing the localization problem in a similar setting (see [5], [7], where localization from angle measurements with bounded errors is performed).

B. Dynamic Setting

In the following set of experiments, the vehicle follows a given trajectory inside a square room. The dynamic model $p(k+1) = p(k) + u(k) + w(k)$ is considered for the position of the vehicle, with $k = 0, \dots, 15$. The driving input $u(k)$ describes the required trajectory, while the disturbance $w(k)$ is assumed to satisfy (12), with constant bound $\varepsilon_i^w(k) = 0.2$, $i = 1, 2, \forall k$.

A first set of experiments is performed with five landmarks in the room and visual angles corrupted by bounded noise, with absolute noise bound equal to 5 degrees. A typical run employing boxes or parallelotopes as approximating regions is reported in Fig. 12. Fig. 13 shows the time variation of uncertainty set areas $\mathcal{R}(k|k)$ and position errors $\|p(k) - \hat{p}(k)\|$ (where $\hat{p}(k)$ is the center of the approximating box or parallelotope). Here the results are averaged over 100 different noise realizations, for the same trajectory and landmark configuration in the room.

Another set of experiments is performed for the case of 20% relative noise bound (i.e., $\varepsilon_{ij}^v(k) = 0.2\theta_{ij}(k)$ for each visual angle measurement, at each time instant k). This time, eight landmarks in the room are used for dynamic localization. A typical run is reported in Fig. 14, while average uncertainty set areas and position errors are shown in Fig. 15.

Figs. 12–15 show that the set membership localization algorithms are able to remarkably reduce localization error and related uncertainty, also in the presence of disturbances in the vehicle dynamics (the term $G(k)w(k)$ in (7)). It can be observed that parallelotopes generally give better approximations of the feasible position set (especially during transients) with respect to boxes, while the two algorithms show similar performance in terms of nominal position error. The large reduction of uncertainty achieved during the first half of the trajectory in Fig. 14 is due to the fact that the robot reaches a location where most of the visual angles are small and therefore small relative errors affect the visual angle measurements.

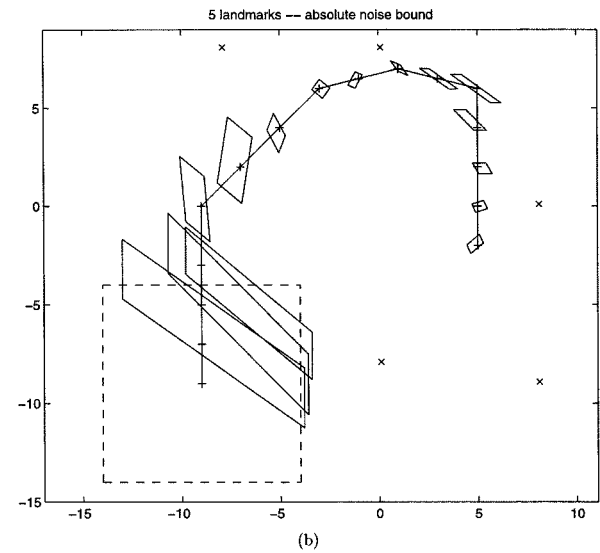
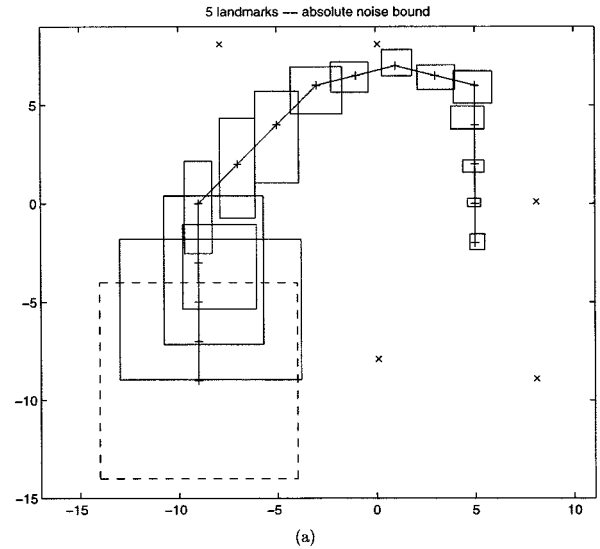


Fig. 12. Dynamic localization with five landmarks and absolute noise bound $\varepsilon^v(k) = 5$. (a) Boxes. (b) Parallelotopes (x: landmarks; solid line: robot trajectory; dashed box: initial feasible position set $\mathcal{L}(0)$).

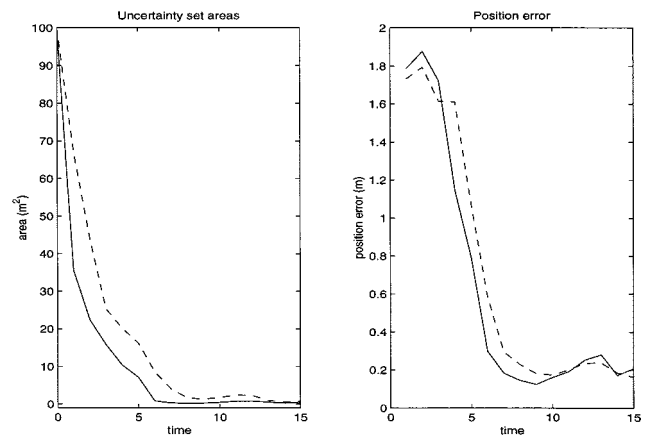


Fig. 13. Average uncertainty set areas and position errors for dynamic localization with five landmarks and absolute noise bound $\varepsilon^v(k) = 5$ (solid line: parallelotopes; dashed line: boxes).

It is worth noting that, due to the approximation introduced by recursive bounding, the size of uncertainty sets can be further

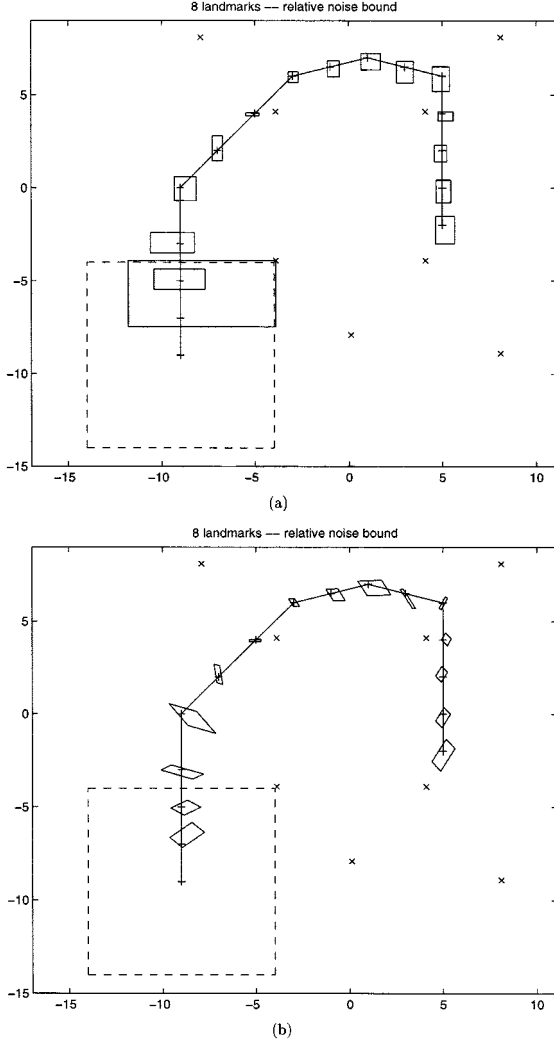


Fig. 14. Dynamic localization with eight landmarks and relative noise bound $\varepsilon_{ij}^v(k) = 0.2\theta_{ij}(k)$. (a) Boxes. (b) Parallelotopes ($\times$: landmarks; solid line: robot trajectory; dashed box: initial feasible position set $\mathcal{L}(0)$).

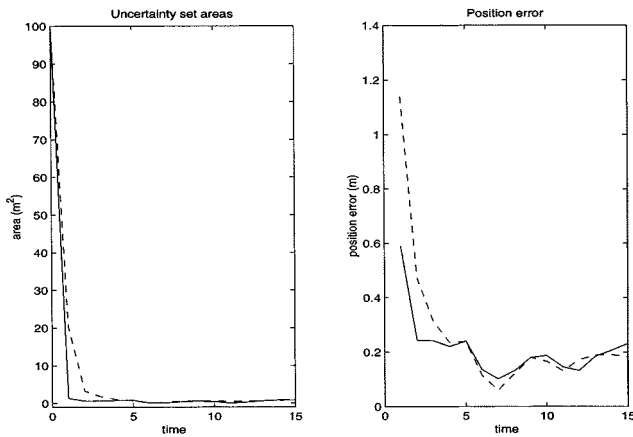


Fig. 15. Average uncertainty set areas and position errors for dynamic localization with eight landmarks and relative noise bound $\varepsilon_{ij}^v(k) = 0.2\theta_{ij}(k)$ (solid line: parallelotopes; dashed line: boxes).

reduced by reprocessing the same measurements several times.

In other words, the quality of set approximation can be trade-off

with the required computational power in order to obtain the maximum uncertainty reduction allowed by the available bandwidth.

VI. CONCLUSION

In this paper, an approach to the localization problem of a mobile robot based on angle measurements has been presented. A set membership framework has been adopted to deal with uncertainty affecting both measurements and the mobile robot dynamics equations. The techniques devised provide recursive estimates for the robot position uncertainty set. These algorithms can be applied successfully in real-time localization and navigation problems, due to the reduced computational complexity of the involved numerical procedures. Several numerical simulation examples show that the proposed approach provides encouraging results.

Although the problem dealt with in the present paper is formulated in a quite simple setting, the set membership approach can be easily extended to include estimation of the robot orientation, landmarks uncertainty and different measurement sources. A more challenging problem, which is a subject of great interest in exploration, concerns the localization problem for the case when the landmarks must also be localized in the considered environment. This is the typical scenario in team-based exploration tasks, where the robots of a subteam act as landmarks for robots of another subteam and all of them concur to the global localization problem and to the construction and updating of the map of the region of interest. Tackling this problem by means of set membership techniques will be the subject of future work.

APPENDIX

Proof of Proposition 1: Let $\Upsilon = \{\xi : \Pi\xi \in \mathcal{M}(k)\}$. First, we show that $\mathcal{R}(k|k-1) \supseteq \Xi(k|k-1)$ and $\mathcal{R}(k|k) \supseteq \Xi(k|k)$. From (20), the claim holds for $k=0$. By induction, let $\mathcal{R}(k-1|k-1) \supseteq \Xi(k-1|k-1)$. From (17) and (21), one has immediately $\mathcal{R}(k|k-1) \supseteq \Xi(k|k-1)$. Then, substituting (22) in (23) and exploiting some straightforward set inclusions, one gets

$$\begin{aligned} \mathcal{R}(k|k) &\supseteq \{[\Pi\mathcal{R}(k|k-1) \cap \mathcal{M}(k)] \times \bar{\Pi}\mathcal{R}(k|k-1)\} \\ &\quad \cap \mathcal{R}(k|k-1) \\ &= \{[\Pi\mathcal{R}(k|k-1) \cap \mathcal{M}(k)] \\ &\quad \times [\bar{\Pi}\mathcal{R}(k|k-1) \cap \mathbb{R}^2]\} \cap \mathcal{R}(k|k-1) \\ &= \{\mathcal{M}(k) \times \mathbb{R}^2\} \cap \mathcal{R}(k|k-1) \\ &\supseteq \Upsilon \cap \Xi(k|k-1) \\ &= \Xi(k|k). \end{aligned}$$

Hence, from (22) one has

$$\begin{aligned} \mathcal{R}_{\Pi}(k|k) &\supseteq \Pi\Xi(k|k-1) \cap \mathcal{M}(k) = \Pi\Xi(k|k-1) \cap \Pi\Upsilon \\ &\supseteq \Pi\{\Xi(k|k-1) \cap \Upsilon\} = \Pi\Xi(k|k) = \mathcal{L}(k) \end{aligned}$$

which concludes the proof.

Proof of Proposition 2: From the definition of box, one has

$$\begin{aligned}
& A(k-1)\mathcal{B}(b, c) + B(k-1)u(k-1) \\
& + G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}\mathcal{B}_\infty \\
& = \{\xi \in \mathbb{R}^4 : \xi = A(k-1)c + A(k-1)\text{Diag}\{b\}\alpha \\
& + B(k-1)u(k-1) + G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}\beta, \\
& \alpha \in \mathbb{R}^4, \beta \in \mathbb{R}^2, \|\alpha\|_\infty \leq 1, \|\beta\|_\infty \leq 1\} \\
& = \{\xi \in \mathbb{R}^4 : \xi = A(k-1)c + B(k-1)u(k-1) \\
& + [A(k-1)\text{Diag}\{b\} G(k-1)\text{Diag}\{\varepsilon^w(k-1)\}]\gamma, \\
& \gamma \in \mathbb{R}^6, \|\gamma\|_\infty \leq 1\}.
\end{aligned}$$

The result easily follows by maximizing $|\xi_i|$, for $i = 1, \dots, 4$, over $\|\gamma\|_\infty \leq 1$.

Proof of Proposition 3: By definition, $\mathcal{R}_\Pi(k|k) \times \bar{\Pi}\mathcal{R}(k|k-1)$ is a box in $\mathbb{R}^4$. Moreover, from (22) it follows that $\mathcal{R}_\Pi(k|k) \subseteq \Pi\mathcal{R}(k|k-1)$ (as $\Pi\mathcal{R}(k|k-1)$ is a box in $\mathbb{R}^2$). Hence,

$$\mathcal{R}_\Pi(k|k) \times \bar{\Pi}\mathcal{R}(k|k-1) \subseteq \mathcal{R}(k|k-1).$$

Therefore,

$$\begin{aligned}
\mathcal{R}_\Pi(k|k) \times \bar{\Pi}\mathcal{R}(k|k-1) &= \mathcal{R}_\Pi(k|k) \\
&\times \{q\xi BBR^2 : q = \bar{\Pi}c \\
&+ \text{Diag}\{\bar{\Pi}b\}\alpha, \|\alpha\|_\infty \leq 1\}
\end{aligned}$$

from which the result immediately follows.

Proof of Proposition 4: Let $p = [x \ y]'$ and consider the computation of

$$\max_{p \in \mathcal{B} \cap \mathcal{C}_{ij}} x \quad (41)$$

(the same reasoning can be repeated for the corresponding minimization problem and for the other coordinate y). Since $\mathcal{E} \subset \mathcal{B} \cap \mathcal{C}_{ij}$, one has

$$\max_{p \in \mathcal{E}} x \leq \max_{p \in \mathcal{B} \cap \mathcal{C}_{ij}} x.$$

Conversely, notice that $\mathcal{B} \cap \mathcal{C}_{ij}$ is a compact set (or possibly the union of disjoint compact sets). Therefore, the maximum of the linear functional x is assumed on the boundary of $\mathcal{B} \cap \mathcal{C}_{ij}$, which contains segments of $\delta\mathcal{B}$ and arcs of circumferences of $\delta\mathcal{C}_{ij}^+$ and $\delta\mathcal{C}_{ij}^-$. The contact points of these segments and arcs can be vertices of $\mathcal{B}$ (points of $\mathcal{E}_1$), intersections of the boundaries of $\mathcal{B}$ and $\mathcal{C}_{ij}$ ($\mathcal{E}_2$), or the landmarks contained in $\mathcal{B}$ ($\mathcal{E}_3$). If the boundary of $\mathcal{B} \cap \mathcal{C}_{ij}$ does not contain arcs of $\delta\mathcal{C}_{ij}^-$, then $\mathcal{B} \cap \mathcal{C}_{ij} \subset \text{Co}(\mathcal{E}_1 \cup \mathcal{E}_2 \cup \mathcal{E}_3)$ and hence

$$\max_{p \in \mathcal{B} \cap \mathcal{C}_{ij}} x \leq \max_{p \in \mathcal{E}_1 \cup \mathcal{E}_2 \cup \mathcal{E}_3} x. \quad (42)$$

Otherwise, if the inequality (42) does not hold, the maximum in (41) is assumed on a point of $\delta\mathcal{C}_{ij}^-$ which does not belong to $\delta\mathcal{B}$. This clearly leads to

$$\max_{p \in \mathcal{B} \cap \mathcal{C}_{ij}} x = \max_{p \in \mathcal{E}_4} x. \quad (43)$$

Hence, from (42) and (43)

$$\max_{p \in \mathcal{B} \cap \mathcal{C}_{ij}} x \leq \max_{p \in \mathcal{E}} x.$$

which completes the proof.

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